

SMART CROP DISEASE DETECTION AND ACTION

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Abstract—Farmers primarily rely on manual inspection to identify crop disease, but this is an intensive process that is often inefficient because it requires expertise. With early symptoms sometimes barely detectable, it can become inaccurate. Written automated disease detection is able to identify many diseases, but results are usually sending to the phone or hardware. This doesn't fit within a farm or crop field real time response or an in-field response to initiate an automated response. In addition to the characteristics noted above can delay the management of the disease, if no action to manage a disease is triggered it can lead to reduced crop yield and increased amounts of pesticides results in an economic loss for a farmer. This work introduces a Smart Crop Disease Detection and Action System - utilizing an AI based image recognition and rapid communication technology. The use of a Convolution Neural Network (CNN) mode of image recognition to recognize diseases such as blight and rust using live captures of leafy images taken from a camera module. The use of an ESP32 will communicate wirelessly and provide triggers for automation. When it detects disease with high confidence, it will send notices/alerts to farmers. The alerts will include the disease name, a captured image, and the confidence level based on the image. Then, once formatted for the user interface, the system initiates automated actions according to the specific disease type. For instance, the system may activate the pesticide spraying module or notify the nearest agricultural extension services. The whole system and actions will be secure, meaning only approved users can reset, turn on and off the overall system, or control the module. GPS will be used to tag the location of the farms, and predictive analytics will utilize information to predict the future outbreak of disease based on past disease information in a specific area, which will allow for future preventive planning efforts. This solution improves the protection of crops, lessens crop loss, and helps in sustainable farming through rapid and automated responses to diseases in the field for broad acre farming or small

agricultural areas.

Index Terms—Smart agriculture; crop disease detection; CNN; IoT; ESP32; real-time monitoring; predictive analytics; automation.

I. INTRODUCTION

Agriculture sustains the global economy and is the major source of food, yet it is subject to crop diseases which dominate the yield quality and quantity. The early detection of plant diseases is the only way to prevent the twofold loss, but farmers often depend on the old manual method which is subjective, laborious, and highly inefficient for vast farms. Furthermore, changes in the environment such as humidity, temperature, and rainfall have all contributed to the quicker spread of crop pests. These hurdles demand the introduction of efficient monitoring systems that are very advanced, highly automated, and extremely real-time to help farmers take corrective actions on time. The recent advancements in IoT, image processing, embedded systems, and machine learning provide the chance to bring about this change by making agriculture solutions that are precise and smart.

The Smart Crop Disease Detection and Action System proposed here is a solution that brings together environmental sensing, the classification of the disease via images, and the automated response mechanisms into one unit. This system uses the sensor modules to take measurements of the most important elements such as temperature, soil moisture, humidity, and light intensity, which are the early signs of the occurrence of the disease constantly. The image of the crop

leaf is captured by a camera module and is processed by a lightweight convolutional neural network (CNN) that has been trained to detect disease symptoms with great accuracy. This technique lessens reliance on manual checks and allows for data-based disease identification that is uniform and consistent over different types of crops. For boosting the performance of

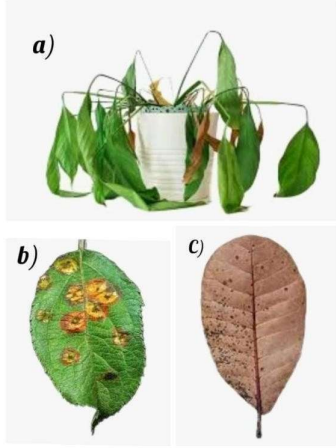


Fig. 1. a.Wilt b.Blight c.Rust

the system, the proposed model automatically takes suitable measures once a disease or an abnormal environmental condition is identified. The actions consist of activating pesticide sprayers, starting irrigation systems, or sending alert notifications to farmers via IoT-based communication channels. The system is also capturing sensor and image data, which will allow farmers to conduct trend analysis, keep monitoring the disease, and come up with agricultural decisions that are based on better information. By the combination of real-time monitoring, intelligent disease detection, and automated control operations the system will a lot to crop protection, will cut down economic losses, and will be promoting sustainable farming practices.

Proposed solution as a whole gives a framework that is scalable, reliable, and cost-effective suitable for modern precision agriculture. It shows how embedded smart technologies and machine learning can change the traditional farming scenario into a more efficient, autonomous, and resilient system.

II. LITERATURE SURVEY

Research in smart agriculture has significantly advanced through IoT-driven monitoring systems, deep-learning-based plant disease analysis, and autonomous data-collection platforms. Agrawal et al. [1] demonstrated how integrating IoT sensing with predictive modeling and reinforcement-learning-based control can optimize environmental conditions inside greenhouses, moving from passive monitoring to active decision-making. In the context of pest management, Prakash and Sungeetha [2] developed a pest categorization and recognition system with associated remedial actions, highlighting the role of intelligent advisory systems for real-time field conditions. Raza et al. [3] explored multi-agent UxV swarms

for fruit disease instance segmentation, showing how autonomous systems can improve both coverage and granularity in agricultural inspections.

Broader analyses of crop data processing techniques have been reported in survey form. Johnson et al. [4] reviewed deep learning architectures for crop analytics and emphasized the need for improved dataset quality, transparency, and multi-modal fusion. For disease diagnosis, Singh et al. [5] proposed hybrid feature extraction—combining handcrafted and learned representations—to improve classification robustness. Gupta et al. [6] employed CNN-based architectures for crop disease prediction, reflecting the continued impact of convolutional models in agricultural imaging.

Advances in sensing and decision support were also explored through multi-sensor integration. Reddy et al. [7] presented an AI-driven crop yield and health monitoring system capable of real-time assessment using IoT data. Akhter and Sofi [8] focused on rice disease diagnosis supported by IoT and machine learning, emphasizing crop-specific feature modeling. For lightweight, on-device plant disease detection, Supriya et al. [9] leveraged a MobileNet architecture in an Android application, enabling offline and low-latency disease prediction suitable for field deployment.

IoT-driven pest and disease management continues to expand into real-time intervention. Kumaran et al. [10] designed a crop pest control system using IoT-enabled monitoring and automated responses, ensuring rapid mitigation in dynamic field environments. Robotic solutions have also emerged: Nagaraj et al. [11] introduced an IoT-based leaf disease identification robot capable of autonomous navigation and detection, demonstrating the potential for fully automated scouting. To address overfitting and dataset limitations, Hafez et al. [12] proposed a deep CNN with L1 regularization for plant disease classification, aiming to improve generalization on challenging agricultural datasets.

Collectively, these works illustrate a clear progression from conventional monitoring to intelligent, autonomous, and data-driven agricultural systems. IoT platforms increasingly support real-time decisions [1], [7], [8], [10]; deep-learning methods continue improving classification performance [5], [6], [9], [12]; and robotics and swarm systems expand the reach of automated field analytics [3], [11]. Despite these advances, challenges remain in achieving robust generalization across environments, enabling seamless multi-sensor fusion, and translating detection outputs into reliable field-level treatments.

III. OBJECTIVE

- To create a continuous monitoring system in real-time that measures temperature, humidity, and soil moisture all the time to give a precise assessment of crop health conditions.
- To use pictures of leaves to carry out the machine-learning-based disease detection, making the early identification of infections possible with very few manual interventions.

- To connect the automatic control actions like irrigation and pesticide spraying through relay-driven modules thus allowing for timely responses that will cut down crop loss and increase the efficiency of precision agriculture.

IV. MATERIALS AND METHODOLOGY

- ESP32 Development Board - This board is the processing unit for the whole system. It receives input data of temperature, humidity, soil, and tank-level along with output data of image-based disease detection.[1] All the processing, real-time communication, and the activation relay-controlled pump or sprayer are done by the ESP32 in parallel through its dual-core processor and Wi-Fi. [3]The very fast GPIO pins ensure reliability as they quickly read the sensor and actuate the devices.
- DHT11 Temperature & Humidity Sensor - This device is the one that is monitoring the environmental conditions of the immediate area.[4] A change in humidity and temperature is a very crucial warning sign of the coming of fungal or bacterial diseases. [7]The ESP32 is there to support the cutting off of environmental logging and timely the taking of preventive actions by reading the sensor's calibrated digital output.
- Soil Moisture Sensor - The sensor's main function is to assess soil moisture level through a resistive sensing probe. [9]The sensor's digital output allows irrigation control based on pre-set thresholds. When the soil dries, the sensor changes its state thereby signaling the ESP32 to activate the water pump. [2]This technique eliminates the risk of crops getting under-watered and thus brings about the farming process of no desiccation.
- OLED Display (SSD1306)- An OLED display is a real-time graphical representation of the inputs and outputs needed to power the system.[3] The OLED display has a black and white resolution of 128 x 64, as well as an I2C protocol to communicate back and forth between ESP32 and the OLED display so that numerous parameters can be displayed effectively while keeping the wiring map as straightforward as possible.
- 4-Channel Relay Module - The 4-channel relay module connects a variety of high-current devices (electrical pumps, spray motors, etc.) to the[4] ESP32 development board. Because relays are used to isolate electrical devices from microcontrollers, they also serve as protection to the microcontroller from damaging surges of electricity. [12]The 4-channel relay module provides control over the irrigation mechanism and the chemical spraying mechanism based on information received from the ESP32's internal programming.
- Water Pump - The water pump kicks in and supplies the plants with water only in case the soil moisture rating goes below the predetermined level. [5]The water pump operates on power from an external source and the relay module controls the pump by activating and deactivating it. [8]The relay module allows the water pump to be controlled and ensures reliable irrigation even large changes in the electrical load on the pump during its operation.
- Float Level Sensors – [9]These devices are installed in each tank to detect the liquid levels and prevent the pumps or sprayers from operating without liquid. [4]They are essential for the system's safety since they allow the irrigation or spraying only when there is sufficient water or pesticide solution.
- 5V External Power Supply – [5]A reliable power source is provided to relays, pumps, and other peripheral modules by this. Powering high-power components will not force the onboard voltage regulator of ESP32 to its limits, which means better reliability for the whole system.
- DC–DC Buck Converter – [9]It reduces high voltages (like 12V) to a constant 5V output for sensors and control electronics.[11] The system's compatibility with different power sources is thereby increased and the risk of damage through power surge is minimized since the voltage is kept constant throughout the operation.
- The Smart Crop Disease Detection and Action system is based on the [9]combination of embedded system control, real-time environmental sensing, machine learning-based disease classification, and automated actuation. [2]The main goal of the methodology is to keep on monitoring the crop health factors, recognize the disease patterns at an early stage, and carry out the irrigation or pesticide spraying without the need for human intervention.
- The system starts to operate by supplying power and initializing the controller. [4]The ESP32 microcontroller is supplied with a 5V regulated power source, while pumps and other similar devices are powered using a separate high-current supply. [7]When the ESP32 board starts, it initializes all GPIO pins, sets up I2C communication with the OLED display and RTC module, and finally performs a system self-check that is very basic but identifies proper connection of sensors, relays, and tank-level switches.
- The phase of continuous environmental sensing starts right away after the initialization is over.[7] DHT11 sensor records the temperature and humidity of the air at intervals. [11]The two conditions are very important to know the possible existence of fungal and bacterial diseases, particularly in the case of high humidity and increased temperature.[8] In the meantime, the soil moisture sensor determines the amount of moisture in the soil through digital threshold logic thus showing whether the soil is dry or suitably moist.
- In addition to environmental measurements, a real-time time[6] stamping is carried out using the DS3231 RTC module. The system can effectively do accurate time-based irrigation scheduling, event logging, and monitoring of disease detection timelines. [4]The RTC chip is powered by an onboard backup battery, so it continues to operate even during power outages.

Images that are taken through either a USB webcam or an external imaging source are first sent to a processing unit where feature extraction and classification are done. The CNN detects the disease and determines the type of it with the trained accuracy. [1]The classification result is communicated back to the ESP32 either through wireless or serial communication.

- After the environmental parameters, [3]soil condition data, and disease classification outputs have been obtained, the decision-making stage is triggered within the ESP32 controller. [9]A specific control algorithm compares the sensed values with the threshold limits. If the soil moisture content goes below the reference value, the irrigation module is activated.[3] If a disease is detected, the respective pesticide sprayer channel is activated according to the disease category.
- Safety verification is done before performing any actuation process. [5]The verification is conducted using float-level sensors that are mounted in the water tank and chemical tanks. [8]The sensors confirm that liquid is supplied and also prevent dry-run conditions that could lead to pump and sprayer damage.[11] The system only activates when adequate tank levels are verified.
- The actuation phase is carried out with the help of a 4-channel relay module, which gives the low-voltage ESP32 Controller and the high-current pump loads complete electrical isolation. In case dry soil conditions are found, the relay turns ON the irrigation pump or activates the corresponding pesticide sprayer according to the disease classification output. The pump runs till the required soil moisture or spraying duration is achieved.
- Throughout the actuation phase,[6] continuous feedback from tank level sensors and moisture sensors guarantees a controlled and safe operation. When the required irrigation or spraying cycle is completed, the ESP32 will turn OFF the relays automatically to save water or avoid chemical over-application.
- A OLED display shows [8]all real-time sensor readings, disease detection results, system decisions, and actuation states. The display is providing continuous monitoring of temperature, humidity, soil moisture, tank status, pump activity, and disease alerts that allow direct field-level monitoring.
- Moreover,[12] the system can send immediate alerts to distant users by making use of the Wi-Fi interface that the ESP32 has inbuilt. [2]Notifications concerning the diagnosis of the disease, the tank level being low, irrigation triggering, and faults within the system can be routed to a mobile app or cloud server for supervising remotely.
- The system keeps on going through the different stages of sensing, detecting, making decisions, and actuating in a continuous loop, thus establishing an automated precision agriculture framework that is closed-loop.[7] This not only assures high reliability but also quick response times, thus making manual field inspections less necessary.

V. PROPOSED SYSTEM

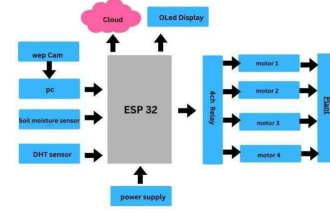


Fig. 2. System Architecture

The Smart Crop Disease Detection and Action System proposed is an IoT-enabled automated agricultural monitoring platform that is meant to supersede slow and manual crop inspection methods. The system makes use of environmental sensing, image-based disease detection, and automated control based on an ESP32 microcontroller. The DHT11 sensor together with a digital soil moisture sensor keep track of the atmospheric and soil conditions, while a USB webcam takes pictures of the crop leaves for visual disease analysis. A trained Convolutional Neural Network (CNN) model classifies diseases like Wilt, Blight, and Rust with very high accuracy.[9] The performance of the CNN model in classification is assessed through the accuracy metric provided by:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

where TP, TN, FP, and FN stand for true positive, true negative, false positive, and false negative values, respectively.

The decision rule is applied to validate the disease detection result when the detection confidence surpasses the pre-set threshold (for instance, 90 percentage).

$$D = \begin{cases} 1, & \text{if } C \geq C_{th} \quad (\text{Disease Detected}) \\ 0, & \text{if } C < C_{th} \quad (\text{Healthy Crop}) \end{cases} \quad (2)$$

Where c stands for the confidence score of the CNN model and C_{th} stands for the pre-set threshold for confidence.

The irrigation decision model based on the threshold is used to determine the soil moisture status:

$$M = \begin{cases} 1, & \text{if } S < S_{th} \quad (\text{Irrigation ON}) \\ 0, & \text{if } S \geq S_{th} \quad (\text{Irrigation OFF}) \end{cases} \quad (3)$$

where S is the measured soil moisture value and S_{th} is the preset moisture threshold. The ESP32 processes both disease detection output and environmental sensor data and performs intelligent decision-making using embedded control logic. The relay activation condition is governed by the logical expression:

$$R = M \vee D \quad (4)$$

In this case, R stands for the relay activation state, M represents the irrigation condition, while D indicates the disease detection condition. Following such reasoning, [9]the controller uses the 4-channel relay module to turn on either the irrigation pump or the right pesticide sprayer. Float-level is

pivotal role in safety verification: they inform the system about the presence of enough water or pesticide solution and only after that the pump can be activated to avoid dry-run damage.

A small OLED display of 0.96 inches shows all the information in real-time about the temperature, humidity, soil moisture, disease detection, pump status, and tank availability. Additionally, the system is capable to send control alerts and event notifications via the ESP32's Wi-Fi connection to a local monitoring unit or mobile dashboard which is close by. The DS3231 Real-Time Clock (RTC) guarantees perfect time stamping of sensor readings, actuation cycles, and system events that makes structured data logging and performance analysis possible.

The electric power equation is used to estimate the power consumption of the system while it is operating:

$$P = V \times I \quad (5)$$

where P is the power consumed in watts, V is the supply voltage, and I is the load current.

The proposed solution works as a closed-loop autonomous agricultural protection system that includes sensing, intelligence, safety validation, and actuation. [8] Consequently, the system no longer requires manual inspection and can carry out disease control instantly; thus, it greatly minimizes crop destruction, chemical use, and also dividends in agriculture. Its modular[9] structure permits installation onsite in small farms, large agricultural areas, and greenhouses at minimum expense and high reliability at the same time.

A. Work Flow

- The process starts with a USB webcam that continuously monitors leaves of the crops while the environmental sensors such as [5]DHT11 temperature and humidity sensor and digital soil moisture sensor are collecting real-time data from the field at the same time.[8] The system, during the normal and healthy conditions, is in an idle monitoring state.
- A trained CNN-based disease detection model, [3]which is deployed on an edge-processing interface coupled with the ESP32, is used to process the images of leaves captured. [12]The model examines and distinguishes between diseases of crops like Wilt, Blight, and Rust by analyzing texture, color distribution, and spot irregularities.
- When the disease classification confidence surpasses the defined threshold value (for instance, 90 percentage),[5] the system performs automatic validation of detection and progresses to alert and control preparation stages.
- The ESP32 microcontroller uses the output of the disease classification together with the readings from the environmental sensors to make a decision about what corrective action to take according to the predefined logic.
- When the moisture in the ground is less than the threshold, [7]the irrigation activity is chosen, and the water pump for that activity is turned on via the relay channel.
- In case a disease is found, the correct pesticide sprayer pump is chosen according to the type of disease recog-

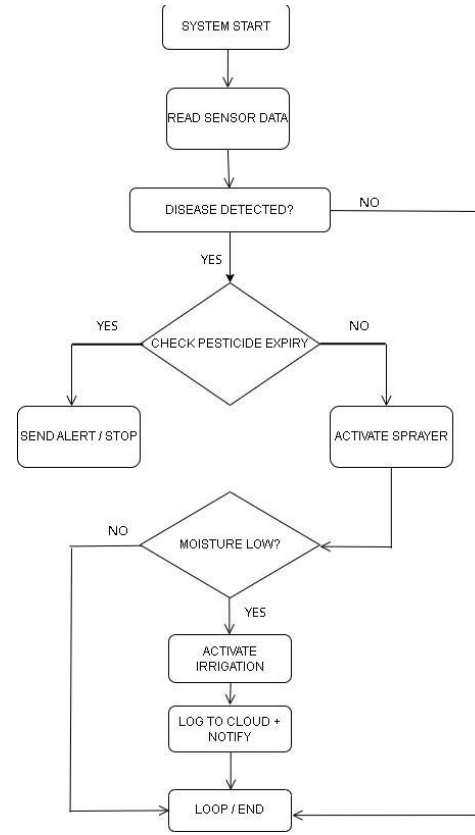


Fig. 3. Workflow

nized by the CNN model, and it is made to operate by the relay module.

- The float-level sensors placed in the water and pesticide tanks check for the[4] sufficient amount of liquid to prevent dry-run conditions and protect the pumping mechanism before the pump is activated.
- The operation for irrigation or spraying is done for a predetermined[5] period while the tank levels and moisture conditions are being monitored all the time to secure controlled application.
- System parameters in real-time such as temperature, humidity, soil moisture, [7]disease detection status, pump activity, and tank availability are shown on the 0.96-inch OLED display.
- The user monitoring interface receives event notifications and system alerts from the [9]ESP32 through its integrated Wi-Fi communication.
- During the process of irrigation or pesticide spraying cycle completion, the ESP32 expires the relay channels by itself and records the operation with time information received from the DS3231 RTC module.
- Crop protection is ensured as the system goes back to its continuous monitoring [7]mode and the overall sensing, de- tection, decision, and actuation cycle is repeated without interruption.

VI. RESULT AND DISCUSSION

The Smart Crop Disease Detection and Action System experiments revealed that the integration of deep learning with IoT-based automation is one of the best practices in precision agriculture. The CNN model trained with leaf images of crops infected by Wilt, Blight, and Rust reached classification accuracies from 70 to 95 percentage, depending on the training dataset. CNN's performance was evaluated according to the accuracy formula given in Eq. (1). Initially, the leaf images acquired using a USB webcam were resized to 224×224 pixels and normalized before being fed to the CNN. By the time the convolutional layers had finished processing, they could draw spatial features like the changes in texture, color distribution, and edge pattern plus the pooling layers had already reduced the attributes and made the features more robust. Through the softmax activation layer, the last classification was done which provided each of the disease classes with probability scores.

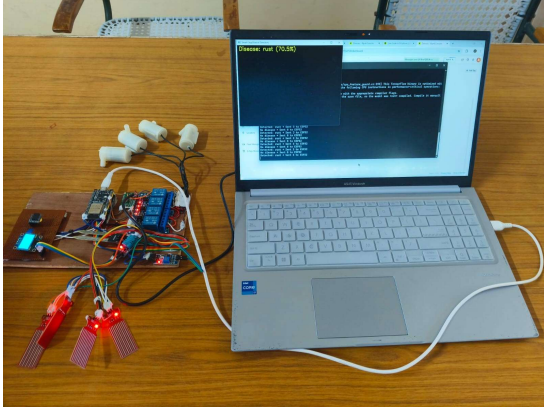


Fig. 4. Hardware Prototype of the Proposed System

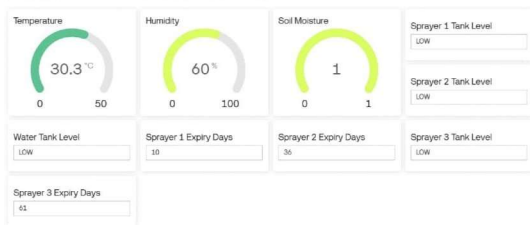


Fig. 5. IoT Monitoring Dashboard

The predicted disease label along with the confidence score was transmitted to the ESP32, the controller where diseases were confirmed or not according to the threshold-based decision rule specified in Eq. (2). Soil irrigation control was operated by the soil moisture threshold model depicted in Eq. (3). If the moisture level dropped below the threshold, irrigation was automatically turned on. The relay control logic was based on the Boolean activation model represented in Eq. (4), which allowed for either irrigation or pesticide spraying depending on the system's decision. The ESP32 took charge of the relay channel control for the water pump or pesticide

sprayers to function as required. Moreover, float-level sensors guaranteed safe operation by stopping dry-run conditions.

In terms of hardware-software integration, the ESP32 has done a good job in acquiring sensor data, controlling relays, and transmitting real-time alerts via Wi-Fi. The DS3231 RTC was used for accurate time-stamping of all sensing and actuation events, which made structured performance analysis possible. The system's power consumption during operation was determined using the electrical power equation described in Eq. (5).

The proposed system outperformed the traditional manual crop monitoring methods in terms of detection accuracy, speed of response, reduction in pesticide application, less crop loss, and better yield forecasting. The combination of CNN-based predictive intelligence with the automated field-level control creates a smart agriculture solution that is reliable, scalable, and economical.

VII. SYSTEM FEATURES EVALUATION

A. Innovation of the Proposed System

- **Dual-Domain Crop Health Intelligence** – The proposed system stands out by integrating vision-based CNN analysis with environmental sensing in real time, so the disease detection can be done based on visual symptoms and surrounding climatic conditions, as well. This combined intelligence advances the early disease detection and the accuracy of the decision-making tree.
- **Direct Relay-Based High-Power Switching Architecture** – The system opts for a direct relay-based switching strategy rather than the conventional driver modules for the control of irrigation and spraying. Therefore, it cuts down the power loss, limits the number of points where the circuit can fail, and ultimately, guarantees a steady switch even under changing field conditions.
- **Adaptive Disease-Triggered Actuation Logic** – The actuation mechanism is not time-based but condition-dependent, wherein pesticide sprayers are activated only when the CNN confidence surpasses the disease threshold. This adaptive logic not only ensures that chemicals are not applied unnecessarily but also increases safety in the fields.
- **Multi-Level Protection Through Liquid Availability Validation** – A new pre-actuation validation layer is realized with the help of float-level sensors in all tanks. This includes an intelligent verification step prior to pump activation, thereby preventing dry-run damage and prolonging the actuator's lifespan.
- **Edge-Level Disease Processing Without Cloud Dependency** – The CNN inference is performed through an edge processing interface directly connected to the ESP32 which allows to detect diseases with almost no delay while the operation continues even in weak network conditions.
- **Integrated Visual-Actuation Synchronization** – The system merges visual recognition of the disease with an instant mechanical response, thus the transition from

detection to action is done without operator approval in case of emergency crop infection situations.

- Self-Contained Autonomous Field Protection Unit – The ESP32 continuously performs sensing, disease validation, safety verification, and actuation in a self-contained closed control loop which makes the whole agricultural protection process completely autonomous and not reliant on manual oversight.

TABLE I
COMPARISON BETWEEN PROPOSED AND EXISTING SYSTEMS

Feature	Existing Systems	Proposed System
Disease Detection Method	Manual leaf inspection by farmers	AI-based CNN model for automatic image detection
Accuracy	Low accuracy due to human error	High accuracy (70–95%)
Time Required	Slow—detected only after visible damage	Fast real-time detection
Action / Response	Manual pesticide/water application	Automatic relay-controlled sprayer/pump
Monitoring Method	No continuous monitoring	Camera + optional sensors
Communication System	No remote alerts	Telegram real-time notifications
Hardware Involvement	Manual tools only	ESP32, camera, relay, sensors
Power Consumption	High manual effort	Low-power microcontroller
Data Logging	Not available	RTC-based time-stamped logging
Environmental Impact	Over-use of pesticides possible	Controlled spraying → less usage
Human Intervention	Fully manual	Minimal human intervention
Reliability	Depends on farmer experience	Stable AI-based performance
Cost	High crop loss increases cost	Reduces loss → cost-effective

Key Advantages Over Existing Models - Design Resilient to the Climate – Incorporates weatherproof sensors and enclosures to guarantee accurate operation in rain, dust, and extreme temperature, unlike many other existing systems that have limitations associated with outdoor conditions. Automatic Prevention of Overflow – Built-in lid closure at 100 percentage capacity prevents spillage and littering, a feature that rarely exists in other smart bin systems.

Safe Access Control: Users authenticate access to open the bin only if they have the passcode, eliminating illegal disposal and use of trash.

VIII. CONCLUSION

By integrating CNN-based disease recognition with IoT-enabled automated control, the researchers have carried out the design, and real-time implementation of a Smart Crop Disease Detection and Action System. The experimental results confirmed the unwavering operation of sensing and actuation units based on ESP32, live environmental monitoring, and correct disease recognition using camera input in a real field scenario. The irrigation and pesticide spraying, which were done automatically according to sensor thresholds and CNN confidence, proved the system to have a closed-loop control capability and real-time responsiveness.

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